

ECE 220 Computer Systems & Programming

Lecture 3 — The Stack

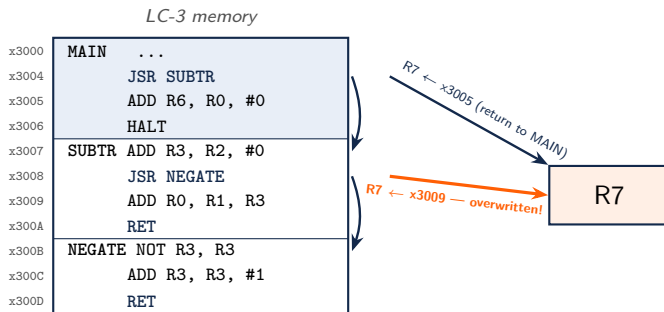
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Electrical & Computer Engineering
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Problem with Saving Register Values in Nested Subroutine Calls

One R7, many nested calls — where do all the return addresses go?



Not unique to R7 — same problem with saving *any* register values, if a subroutine calls itself.

Approach: Storage that Grows with Each Call, in Reverse Order

1. Which data structure returns things in exactly *reverse* order?
2. How do we build one from plain memory and a single register?

Today: the stack — one of the most used data structures in computing.
Yes, *that* stack: stackoverflow.com is named after the error you get when nested calls run it out of room.

Approach: Storage that Grows with Each Call, in Reverse Order

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Level 1: rainy city (Yusuf's dream)



Level 2: the hotel (Arthur's dream)



Level 3: snow fortress (Eames's dream)

Inception (C. Nolan, 2010), © Warner Bros.

Warm-up: Would This Program Work?

```
.ORIG x3000
    AND R1, R1, #0
    AND R2, R2, #0
    ADD R1, R1, #5    ; R1 = 5
    ADD R2, R2, #2    ; R2 = 2
    JSR SUBTR        ; R0 <- R1 - R2
    ADD R6, R0, #0    ; copy result
    HALT
```

```
; SUBTR: R0 <- R1 - R2
SUBTR  ADD R3, R2, #0
        JSR NEGATE        ; R3 = -R2
        ADD R0, R1, R3    ; R0 = R1 - R2
        RET
; NEGATE: R3 <- -R3
NEGATE NOT R3, R3
        ADD R3, R3, #1
        RET
```

Work with a neighbor: does the program reach **HALT**?

What is in R0?

Fix: SUBTR must save R7 before its own JSR — but saving to labeled slots can't handle *arbitrary* nesting. We need something better.

Stack — An Abstract Data Type

A **LIFO (last-in, first-out)** storage structure:

- The **first** thing you put in is the _____ thing you take out
- The **last** thing you put in is the _____ thing you take out

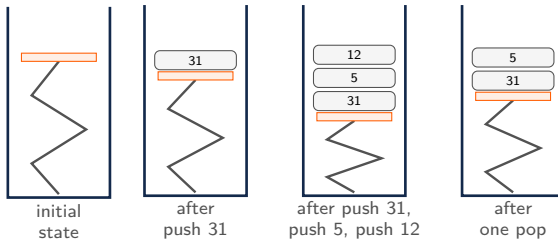
The **means of access** is what defines a stack — not the implementation.

- **Push**: add an item to the top
- **Pop**: remove the item at the top
- **IsFull**: a push when full is an _____
- **IsEmpty**: a pop when empty is an _____

What else in computing behaves like a stack?

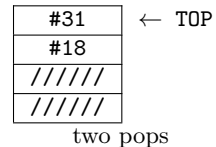
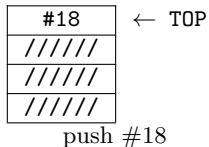
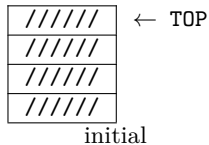
A Physical Stack: the Coin Holder

First coin in is the **last** coin out.



Implementation in Hardware: Data Moves

Data items move; the top of the stack is a fixed location.



Fine for hardware (shift registers) — wasteful in memory: every push would copy every element.

Implementation in Memory: the Pointer Moves

Data items stay put; only our idea of where the top is moves.

x3FFB	/////
x3FFC	/////
x3FFD	/////
x3FFE	/////
x3FFF	/////

R6: x4000 ← TOP
initial

x3FFB	/////
x3FFC	/////
x3FFD	/////
x3FFE	/////
x3FFF	#18

R6: x3FFF ← TOP
push #18

x3FFB	/////
x3FFC	#12
x3FFD	#5
x3FFE	#31
x3FFF	#18

R6: x3FFC ← TOP
push #31, push #5, push #12

x3FFB	/////
x3FFC	#12
x3FFD	#5
x3FFE	#31
x3FFF	#18

R6: x3FFE ← TOP
two pops

- By convention, **R6** holds the **Top-of-Stack (TOS)** pointer
- The stack **grows toward lower addresses**

Basic Push and Pop

With R6 pointing at the top element:

Push (R0 holds the value):

_____ ; make room (grow downward)
_____ ; store at the new top

Pop (value comes back in R0):

_____ ; read the top
_____ ; shrink the stack

Two instructions each — but what happens if we push when full, or pop when empty?

A Safe PUSH: Setup

x3FF0		← STACK_END
⋮		
x3FFE		
x3FFF		
x4000		← STACK_START

16 usable slots: x3FF0–x3FFF.

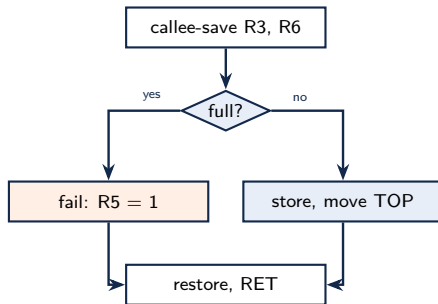
`STACK_TOP` lives in *memory* — the subroutine loads it into R6, uses it, and stores it back.

```
STACK_START .FILL x4000
STACK_END   .FILL x3FF0
STACK_TOP   .FILL x4000    ; empty: TOP = START
P_SAVER3    .BLKW #1
P_SAVER6    .BLKW #1
```

The contract:

- **IN:** R0 = value to push
- **OUT:** R5 = 0 (success) or 1 (_____)
- Everything else: callee-saved
- Full when TOP = _____

In-Class Problem: PUSH with Overflow Checking



How do we compare TOP with
STACK_END?

```

PUSH    ST    R3, P_SAVER3
        ST    R6, P_SAVER6
; init R5 = 0 (assume success)

; R3 <- STACK_END, R6 <- STACK_TOP

; full when TOP - END == 0 -> P_OVER

; else: decrement TOP, store R0, save TOP

        BR    P_DONE
P_OVER  ; set R5 = 1 (overflow)

P_DONE  LD    R3, P_SAVER3
        LD    R6, P_SAVER6
        RET
  
```

A Safe POP: Setup

x3FF0		← STACK_END
⋮		
x3FFE		
x3FFF		
x4000		← STACK_START

Underflow: popping from an *empty* stack
— there is no top element to read.

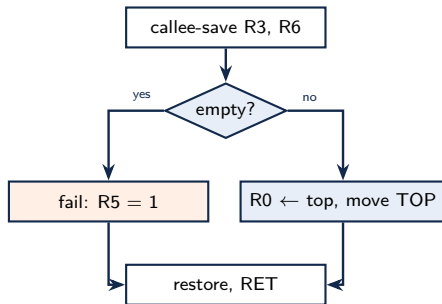
Same discipline as PUSH: load `STACK_TOP`
into R6, use it, store it back.

```
STACK_START .FILL x4000
STACK_END   .FILL x3FF0
STACK_TOP    .FILL x4000    ; empty: TOP = START
Q_SAVER3     .BLKW #1
Q_SAVER6     .BLKW #1
```

The contract:

- **IN:** nothing
- **OUT:** R0 = _____; R5 = 0 (success) or 1 (_____)
- Everything else: callee-saved
- Empty when TOP = _____

In-Class Problem: POP with Underflow Checking



How do we detect *empty*?

```

POP      ST    R3, Q_SAVER3
         ST    R6, Q_SAVER6
         ; init R5 = 0 (assume success)

         ; R3 ← STACK_START, R6 ← STACK_TOP

         ; empty when TOP - START == 0 -> Q_UNDER

         ; else: R0 ← top value, increment TOP, save TOP

         BR    Q_DONE
Q_UNDER  ; set R5 = 1 (underflow)

Q_DONE   LD    R3, Q_SAVER3
         LD    R6, Q_SAVER6
         RET
  
```

Summary & What's Next

Today

- **The problem:** nested calls need somewhere safe to put return addresses and saved registers
- **The stack ADT:** LIFO; push, pop, overflow, underflow — defined by the access rule, not the implementation
- **In memory:** data stays put, **R6** (the TOS pointer) moves; the stack grows toward lower addresses
- **Safe PUSH & POP:** check for overflow/underflow, report success/failure in R5, callee-save everything you touch

Next lecture

- Programming *with* the stack: evaluating arithmetic expressions

Code from today: `nested_bug.asm`, `push.asm`, `pop.asm` in the course code repository.